

M.N.P.



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY

PAPER 3

6031/3

2 hours 30 minutes

NOVEMBER 2024 SESSION

Additional materials:

Answer paper

Data Booklet

Electronic calculator

Graph paper

INSTRUCTIONS TO CANDIDATES

Answer **six** questions.

Write your answers on the separate answer paper provided.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 13 printed pages.

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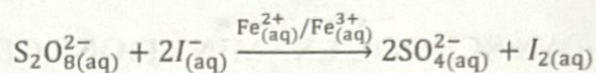
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Section A

Answer any **two** questions from this section.

- 1 (a) The chemical equation for the reaction between peroxodisulphate ions and iodide ions is shown.



The reaction is catalysed by $\text{Fe}^{2+}(\text{aq})/\text{Fe}^{3+}(\text{aq})$.

- (i) Define the term homogenous catalysis.
- (ii) Describe and explain the effect of the catalyst on the reaction.
- (iii) Explain how the rate of the catalysed reaction is affected by increasing the pH of the reaction mixture to a pH of 12.

[5]

- (b) When hydrogen iodide, HI is heated in a transparent closed container, violet fumes are observed and the intensity of the violet colour increases with increasing temperature.

- (i) Write a balanced chemical equation for the decomposition of HI .
- (ii) Explain the observation.

[3]

- (c) Compare the acidity of HF and HI .

[2]



(d) Fig.1.1 shows a fuel cell.

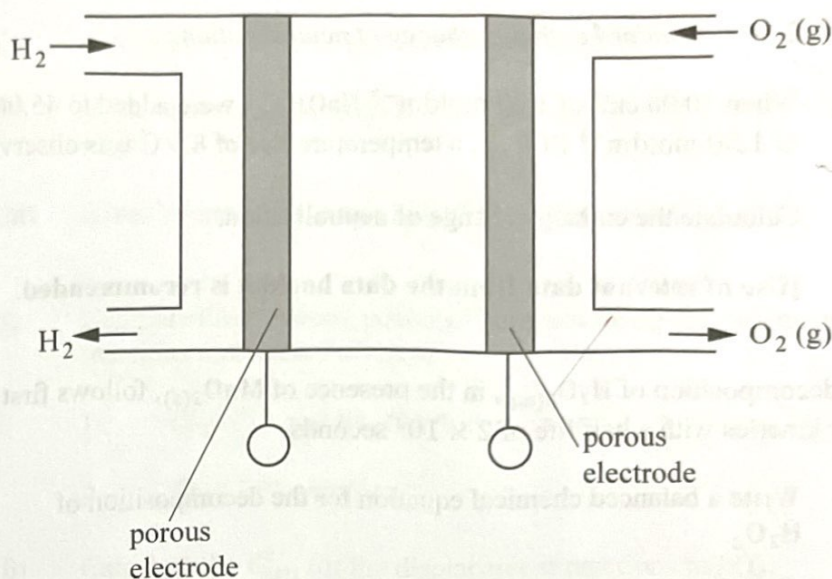


Fig. 1.1

- Define the term *fuel cell*.
- Name the fuel in the cell shown in Fig. 1.1
- Identify the anode in the fuel cell.
- Write balanced chemical equations for the anodic and cathodic reactions.

[5]

[Total:15]

- 2 (a) A volume of 5 dm^3 of $H_{2(g)}$ at 100°C was mixed with 5 dm^3 of $N_{2(g)}$ at 20°C in a flexible balloon at constant pressure. The temperature of the gas mixture was then adjusted to 25°C .

Calculate the final volume of the gas mixture.

Handwritten calculations:

$$Pv = nRT$$

Calculate No. of H_2 & N_2

$$n = \frac{5}{8.31 \times 293} = 2.053531458 \text{ moles}$$

$$n = \frac{5}{8.31 \times 373} = 1.613$$

$$V = 3.666627207 \times 8.31$$

$$V = 9079.96 \text{ m}^3$$

Turn over

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- (b) (i) Define the term

1 *bond energy,*

2 *standard enthalpy change of neutralisation.*

- (ii) When 50.00 cm^3 of $1.20 \text{ mol dm}^{-3} \text{ NaOH}_{(\text{aq})}$ were added to 45.00 cm^3 of $1.50 \text{ mol dm}^{-3} \text{ HCl}_{(\text{aq})}$, a temperature rise of 8.5°C was observed.

Calculate the enthalpy change of neutralisation.

[Use of relevant data from the data booklet is recommended.]

[6]

- (c) The decomposition of $\text{H}_2\text{O}_{2(\text{aq})}$, in the presence of $\text{MnO}_{2(\text{s})}$, follows first order kinetics with a half life of 2×10^2 seconds,

- (i) Write a balanced chemical equation for the decomposition of H_2O_2 .
- (ii) Explain the catalytic effect of $\text{MnO}_{2(\text{s})}$ in terms of the Maxwell Boltzmann distribution curve.
- (iii) Calculate the time required for the concentration of $\text{H}_2\text{O}_{2(\text{aq})}$ to decrease from 0.2 mol dm^{-3} to 0.05 mol dm^{-3} in the presence of $\text{MnO}_{2(\text{s})}$.

[6]

[Total:15]

- 3 (a) (i) State the **three** main points of the Valency Shell Electron Pair Repulsion Theory.

- (ii) Draw a structure of H_2SO_4 , showing bond angles.

[5]

- (b) (i) Draw a labelled diagram of a cell that can be set up by connecting the Mg^{2+}/Mg half cell to the Fe^{2+}/Fe half cell.

- (ii) Calculate the E^θ cell.

[Use of relevant data from the Data Booklet is recommended]

- (iii) Give **one** use of Mg, based on the cell.

[5]

- (c) (i) Draw a dot and cross diagram to show the bonding in the dimer Al_2Cl_6 .

- (ii) Explain why Al_2Cl_3 is able to dimerise.

- (iii) Suggest why bonding in Al_2Cl_3 is covalent while Al_2F_3 is ionic.

[5]

[Total:15]



Section B

Answer **one** question from this section.

- 4 (a) (i) Give all the oxides of phosphorus and sulphur.
- (ii) Write chemical equations for the reactions of each of the oxides in (i) with water.
- (iii) Give the order of the strength of the acids formed. [5]
- (b) (i) Compare the oxidising power of halogens using displacement reactions that occur between:
- 1 $\text{Cl}_2 / \text{Cl}^-$ and $\text{Br}_2 / \text{Br}^-$,
- 2 $\text{Br}_2 / \text{Br}^-$ and I_2 / I^- .
- (ii) Calculate the $E_{\text{cell}}^\ominus$ for the displacement reactions in b(i). [3]
- (c) (i) Describe and explain how the solubility of Group (II) sulphates vary down the group.
- (ii) Write an equation for the thermal decomposition of MgCO_3 .
- (iii) Compare the thermal stabilities of BaCO_3 and MgCO_3 . [7]

[Total:15]

- 5 (a) Phosphorus (V) oxide has a very high affinity for water.
- (i) Give **one** use of phosphorus (V) oxide linked to its affinity for water.
 - (ii) Write a chemical equation to show how phosphorus (V) oxide reacts with water.
 - (iii) Suggest the pH value of the resultant solution in (a)(ii).
 - (iv) Write the equation for the reaction between phosphorus and chlorine.
- [4]
- (b) (i) Give **one** use of iodine in humans.
- (ii) Write the general outer electronic structure of halogens.
 - (iii) Describe, with the aid of an equation, how chlorine is used in water treatment.
- [5]
- (c) Hydrogen bromide reduces sulphuric acid to sulphur dioxide. Hydrogen iodide reduces sulphuric acid to hydrogen sulphide.
- (i) Write equations for the **two** reactions.
 - (ii) Give a reason for the variation in reducing powers of Br^- and I^- .
 - (iii) Compare the acidities of HBr and HI .
 - (iv) Suggest a reason for the differences in acidity.
- [6]
[Total:15]



Section C

Answer any **two** questions from this section.

6 (a) Give any one:

- (i) physical property of arenes.
 (ii) chemical property of arenes, (A)

[2]

(b) The structure of an organic compound, **X**, is shown in Fig. 6.1.

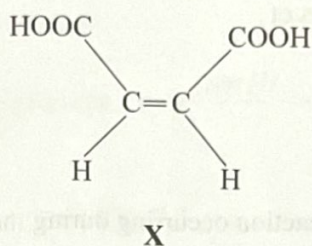
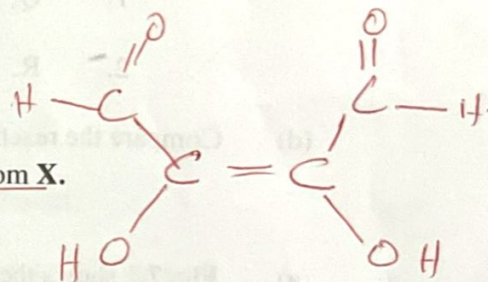


Fig. 6.1

NO of Carbon
functional group
with possible
Alphabetical order

- (i) Give the IUPAC name of **X**.
 (ii) State the types of isomers that can be formed from X.
 (iii) Draw the structures of the isomers of **X**.
 (iv) Deduce the organic products formed when **X** reacts with:



1. hot alkaline $\text{KMnO}_4(\text{aq})$,

It will form CO_2 then the alkaline will react to form salt. in acidic it's CO_2 then in alkaline it's COO^- (ion) salt.

2. LiAlH_4 under reflux.

[7]

There is difference isomers formed from **X** and structure formed ~~for~~ by **X**. $\rightarrow$ geometrical structural
Cis-trans.



- (c) Fig 6.2 shows how phenylamine can be converted to other organic compounds.

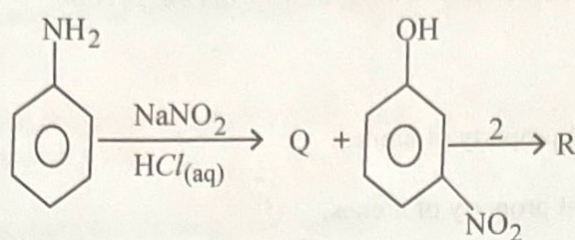


Fig. 6.2

- (i) Draw the structures of

1. Q,
2. R.

- (ii) Give the type of reaction occurring during the formation of

1. Q,
2. R.

[4]

- (d) Compare the reactivity of phenylamine and ethylamine.

[2]

[Total:15]

- 7 (a) Fig. 7.1 shows the structure of methyl orange, an example of an azo compound.

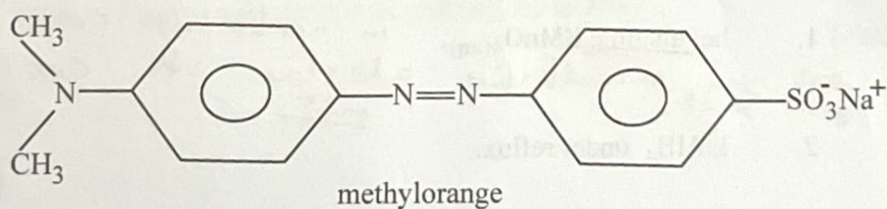


Fig. 7.1

- (i) State the acid/base nature of methyl orange.
- (ii) Suggest, with reasons, a role of the $-\text{SO}_3^- \text{Na}^+$ group.
- (iii) Give any **two** common uses of azo compounds,

[5]



- (b) (i) Fig. 7.2 shows how polystyrene can be synthesised from 1,2-dichloroethane and benzene.

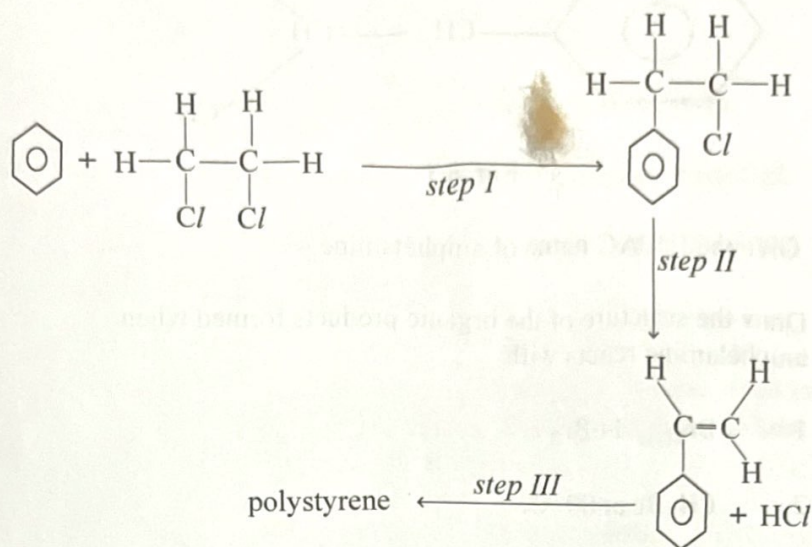


Fig. 7.2

Give the reagents and condition for

1. step I,

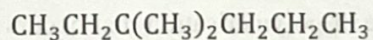
2. step II.

(ii) Draw a section of polystyrene showing two repeat units.

(iii) Name any **two** factors that determine the melting point and strength of polystyrene.

[7]

- (c) The catalytic cracking of the compound shown gives propene as one of the products.



(i) State any **one** condition used in the cracking process.

(ii) Deduce the structural formula of the other product.

(iii) State the importance of the catalytic cracking process.

[3]

[Total: 15]



- 8 (a) Amphetamine, a stimulating drug, has a structure shown in Fig. 8.1.

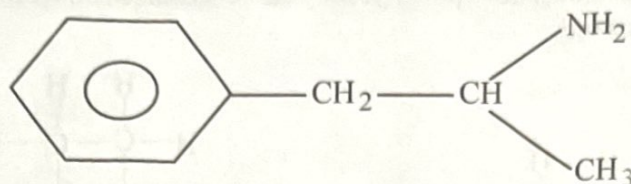


Fig. 8.1

- (i) Give the IUPAC name of amphetamine.
- (ii) Draw the structure of the organic products formed when amphetamine reacts with
 - 1 $\text{Br}_{2(l)} / \text{FeBr}_3$,
 - 2 CH_3Br at 80°C .
- (iii) Name the types of reactions in (a) (ii).

[5]

- (b) (i) Give the main use of alkanes.
- (ii) Explain the unreactivity of alkanes.
- (iii) Methane reacts with chlorine as follows:



Give the mechanism that produces CH_2Cl_2 .

[7]

- (c) Proteins are natural polymers.
 - (i) Explain the term *natural polymer*.
 - (ii) State the type of polymerisation that occurs during protein formation.

[3]

[Total:15]



Section D

Answer **one** question from this section.

9 (a) (i) Give one

1. source of the sulphur contained in fossil fuels,
2. hazard associated with the extraction of fossil fuels.

(ii) Describe, with the aid of equations, the removal of sulphur dioxide using fluidized limestone beds.

[5]

(b) Some medicinal plant tissues collected for analysis were divided into two portions. The first portion was crushed and extracted using hexane. The second portion was crushed and extracted using water.

- (i) State the differences between the two extracts.
- (ii) The extracts were analysed using thin layer chromatography (TLC).

Suggest a suitable mobile phase for the TLC analysis of the

1. hexane extract,
2. water extract.

(iii) Give

1. **two** advantages of using water as a solvent in the extraction,
2. **two** disadvantages of using water as a solvent in the extraction.

[5]

(c) (i) Define the term *ligand*.

(ii) Explain why the colours of $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}_{(\text{aq})}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ are different.

(iii) Explain why NH_3 can displace H_2O from complexes.

[5]

[Total: 15]



10 (a) (i) Describe and explain the observations made when H_2SO_4 is added to K_2CrO_4 .

(ii) Draw diagrams to show isomers of $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$.

(iii) Calculate the oxidation state of Cr in the complex in (ii).

(b) (i) Identify the greenhouse gas produced from

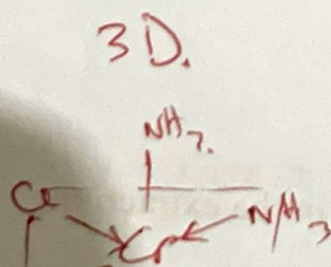
1. landfill sites,

2. incinerators.

(ii) Give

1 two advantages of incineration,

2 two disadvantages of incineration.



- (c) The compounds shown in Fig. 10.1 were analysed using paper chromatography with a non polar solvent.

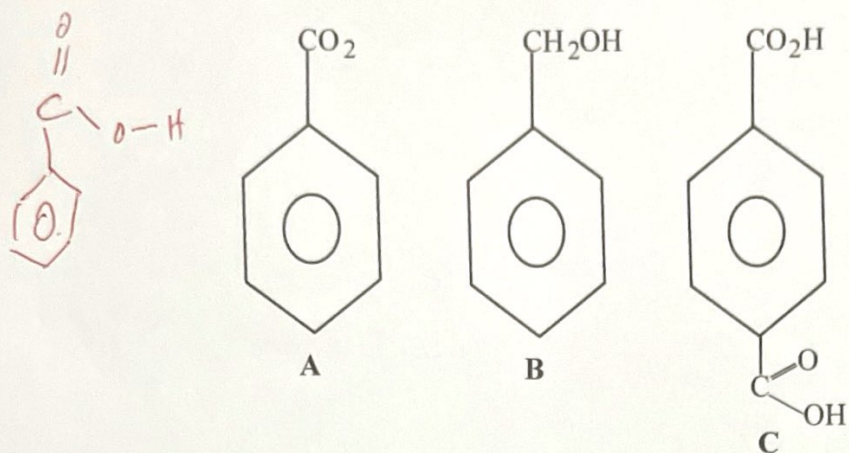


Fig. 10.1

The chromatogram obtained is shown in Fig. 10.2.

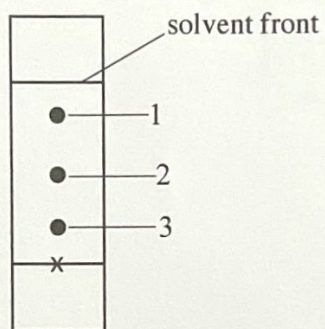


Fig. 10.2

Identify, with reasons, the compound responsible for each spot.

[4]

[Total: 15]

